

Full Stack Development with MERN

Project Documentation

1. Introduction

Project Title

Online Complaint Registration

Team Members

Name	Role
S.Vandana	Frontend Developer
G.Susmitha	Backend Developer
G.Susmitha	Database Developer
S.Vandana	Testing & Documentation

. Project Overview

Purpose

The **Online Complaint Registration** is a web-based application developed using the MERN stack. It enables citizens to register complaints online, upload supporting documents, track complaint status, and receive notifications. Administrators can efficiently manage complaints, assign them to officers, update statuses, and generate reports.

Features

- User Registration & Login
- Secure Authentication (JWT)
- Complaint Submission
- Complaint Tracking
- Upload Complaint Documents
- Admin Dashboard
- Complaint Status Updates
- User Profile Management
- Complaint Reports
- Email Notifications

3. Architecture

Frontend

Technology

- React.js
- HTML5
- CSS3
- Bootstrap
- Material UI
- Axios

Responsibilities

- User Interface
- Login/Register
- Dashboard
- Complaint Form
- Complaint List

- Reports
- Profile

Backend

Technology

- Node.js
- Express.js

Responsibilities

- Authentication
- Complaint APIs
- User APIs
- Admin APIs
- Report Generation
- Email Notifications

Database

Technology

- MongoDB
- Mongoose

Collections

- Users
- Complaints
- Categories
- Reports
- Notifications

4. Setup Instructions

Prerequisites

- Node.js
- MongoDB Community Server
- MongoDB Compass
- Git
- Visual Studio Code

Installation

Step 1

Clone Repository

```
git clone
```

```
https://github.com/yourusername/OnlineComplaintRegistration.git
```

Step 2

Go to project folder

```
cd OnlineComplaintRegistration
```

Step 3

Install Frontend Dependencies

```
cd client
```

```
npm install
```

Step 4

Install Backend Dependencies

```
cd ../server
```

```
npm install
```

Step 5

Create `.env`

`PORT=5000`

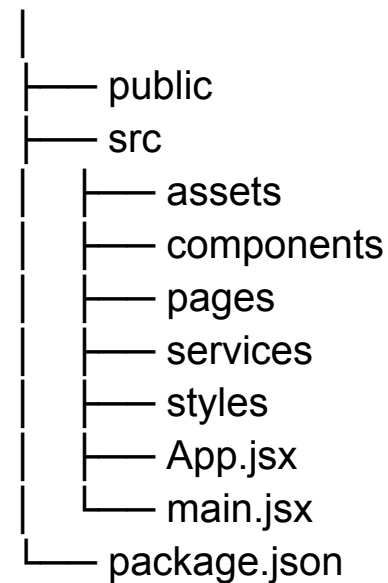
`MONGO_URI=mongodb://localhost:27017/ComplaintDB`

`JWT_SECRET=yourSecretKey`

5. Folder Structure

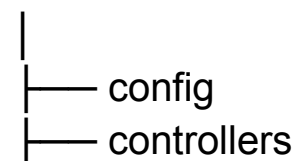
Client

client/



Server

server/





- middleware
- models
- routes
- uploads
- utils
- .env
- server.js
- package.json

6. Running the Application

Start Backend

```
cd server  
npm start
```

or

```
nodemon server.js
```

Start Frontend

```
cd client  
npm run dev
```

Open

<http://localhost:5173>

7. API Documentation

Authentication

Method	Endpoint	Description
POST	/api/auth/register	Register User
POST	/api/auth/login	Login User

Complaints

Method	Endpoint	Description
GET	/api/complaints	View Complaints
POST	/api/complaints	Submit Complaint
PUT	/api/complaints/:id	Update Complaint
DELETE	/api/complaints/:id	Delete Complaint

Users

Method	Endpoint	Description
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GET	/api/users/p rofile	User Profile
PUT	/api/users/p rofile	Update Profile

Admin

Meth od	Endpoint	Descrip tion
GET	/api/admin/das hboard	Dashbo ard
GET	/api/admin/repo rts	Reports

Example Request

```
POST /api/auth/login
{
  "email":"user@gmail.com",
  "password":"123456"
}
```

Example Response

```
{
  "success": true,
  "token":"JWT_TOKEN"
```

}

8. Authentication

The application uses **JWT (JSON Web Token)** for authentication.

Process

- User registers.
- Password is encrypted using **bcrypt**.
- User logs in.
- Server verifies credentials.
- JWT token is generated.
- Token is stored on the client.
- Protected routes verify the token before granting access.

9. User Interface

Include screenshots of:

- Home Page
- Login Page
- Registration Page
- Dashboard
- Complaint Form
- Complaint List
- Profile
- Admin Dashboard
- Reports Page

10. Testing

Testing Strategy

- Unit Testing
- API Testing
- Integration Testing
- Manual UI Testing

Tools

- Postman
- Chrome Developer Tools

11. Screenshots / Demo

Include screenshots of:

- Home
- Login
- Register
- Dashboard
- Submit Complaint
- Complaint List
- Admin Dashboard
- Reports

or

GitHub Repository

<https://github.com/suraparajuvandana/OnlineComplaintRegistration>

12. Known Issues

- Email notification requires SMTP configuration.
- SMS notification is not implemented.
- Large file uploads may take additional time.
- Cloud deployment is pending.

13. Future Enhancements

- AI-based Complaint Classification
- Complaint Priority Prediction
- Mobile Application (Android/iOS)
- SMS Notifications
- Real-Time Chat Support
- Google Maps Integration
- Complaint Analytics Dashboard
- Cloud Deployment (AWS/Render)
- Multi-language Support
- Push Notifications

This documentation is suitable for a **MERN Stack Online Complaint Registration and Management System** and follows the format requested for your project.